

Objective: You are given a CSV file containing monthly sales data for a retail company. Your goal is to perform basic analysis and prepare the data for future dashboard integration.

Data Cleaning – removing duplicate rows and handling missing values.

I checked the dataset, so is the AI. My opinion – everything is ok with the Data, no duplicates and no missing values. AI says the same. The only thing I changed is at the month column – first is the year then the month (2023-03) I am changing their places like this (03.2023). So, I am starting with the Analysis.

The Analysis

1. Calculating total sales per product category

I am using this code in my SQL Workbench:

```
SELECT
product_category,
SUM(sales_revenue) AS total_revenue
FROM
project11.sales_data
GROUP BY
product_category
ORDER BY
total_revenue DESC;
```

And here is the result:

	product_category	total_revenue
►	Clothing	37498.86
	Books	26617.07
	Electronics	25463.31
	Home	15676.39
	Toys	14657.14

TOTAL SALES BY PRODUCT CATEGORY	
PRODUCT CATEGORY	TOTAL REVENUE
<i>Clothing</i>	37498.86
<i>Books</i>	26617.07
<i>Electronics</i>	25463.31
<i>Home</i>	15676.39
<i>Toys</i>	14657.14

According to the information I received after the coding in the SQL, it seems that we have the following classification:

1. *Clothing – first place, biggest profit with 37498.86;*
2. *Books – second place, profit of 26617.07;*
3. *Electronics – third place with profit of 25463.31;*
4. *Home – fourth place with profit of 15676.39;*
5. *Toys – fifth and last place with profit of 14657.14*

2. TOP 5 PRODUCTS BY REVENUE

SQL Code:

```
SELECT
product_name,
SUM(sales_revenue) AS total_revenue
FROM
project11.sales_data
GROUP BY
product_name
ORDER BY
total_revenue DESC
LIMIT 5;
```

Result:

	product_name	total_revenue
▶	T-Shirt	23871.85
	Laptop	11993.93
	Biography	11962.56
	Chair	10478.52
	Novel	10408.56

TOP 5 PRODUCTS BY REVENUE	
PRODUCT	REVENUE
<i>T-shirt</i>	23871.85
<i>Laptop</i>	11993.03
<i>Biography</i>	11962.56
<i>Chair</i>	10478.52
<i>Novel</i>	10408.56

1. *T-shirt* – with profit of 23871.85;
2. *Laptop* – with profit of 11993.03;
3. *Biography* – with profit of 11962.56;
4. *Chair* – with profit of 10478.52;
5. *Novel* – with profit of 10408.56

3. Determining monthly sales trends

I am using the following SQL code to find out the total revenue and total quantity of products sold for each month:

```

SELECT
month,
SUM(sales_quantity) AS total_quantity,
SUM(sales_revenue) AS total_revenue
FROM
project11.sales_data
WHERE
month IN ('2023-01', '2023-02', '2023-03', '2023-04', '2023-05', '2023-06')
GROUP BY
month
ORDER BY
month;

```

Result:

	month	total_quantity	total_revenue
►	2023-01	373	18120.08
	2023-02	801	47186.89
	2023-03	157	12423.18
	2023-04	268	11873.90
	2023-05	260	14046.05
	2023-06	352	16262.67

Month	Total Quantity of Sold Products	Total Revenue
January 2023	373	18120.08
February 2023	801	47186.89
March 2023	157	12423.18
April 2023	268	11873.90
May 2023	260	14046.05
June 2023	352	16262.67

During the first half of 2023 there is much revenue during February, also in January is 373, but not that much and in June is 352. Overall there isn't much stock sold.

Now I am going to check for each month which product brings the biggest profit and which category:

	month	product_category	total_quantity
►	2023-01	Books	228
	2023-01	Electronics	78
	2023-01	Toys	67
	2023-02	Books	244
	2023-02	Clothing	206
	2023-02	Electronics	124
	2023-02	Home	163
	2023-02	Toys	64
	2023-03	Books	82
	2023-03	Clothing	62
	2023-03	Home	13
	2023-04	Books	25
	2023-04	Clothing	96
	2023-04	Electronics	147
	2023-05	Clothing	115
	2023-05	Electronics	79
	2023-05	Home	23
	2023-05	Toys	43
	2023-06	Books	11
	2023-06	Clothing	137
	2023-06	Electronics	72
	2023-06	Home	104
	2023-06	Toys	28

I used the following codes:

For Product Category:

```
SELECT  
  
month,  
  
product_category,  
  
SUM(sales_quantity) AS total_quantity  
  
FROM  
  
project11.sales_data  
  
GROUP BY  
  
month, product_category  
  
ORDER BY  
  
month, product_category;
```

And this is for the Products:

```
SELECT  
  
month,  
  
product_name,  
  
SUM(sales_quantity) AS total_quantity  
  
FROM  
  
project11.sales_data  
  
GROUP BY  
  
month, product_name  
  
ORDER BY  
  
month, product_name;
```

Month	Product Category	Total Quantity
01.2023	Books	228
01.2023	Electronics	78
01.2023	Toys	67
02.2023	Books	244
02.2023	Clothing	206
02.2023	Electronics	124
02.2023	Home	163
02.2023	Toys	64
03.2023	Books	82
03.2023	Clothing	62
03.2023	Home	13
04.2023	Books	25
04.2023	Clothing	96
04.2023	Electronics	147
05.2023	Clothing	115
05.2023	Electronics	79
05.2023	Home	23
05.2023	Toys	43
06.2023	Books	11
06.2023	Clothing	137
06.2023	Electronics	72
06.2023	Home	104
06.2023	Toys	28

- January 2023: We have so many books sold – 228, and not that much Electronics – 78 and Toys – 67. Total is **373** as we already know;
- February 2023: Here is a bigger profit. Books – 244, Clothing – 206. Also Electronics – 124, Home – 163 and Toys – 64. Total = **801**;
- March 2023: Books – 82, Clothing – 62, Home – 13 = **157**;
- April 2023: Books – 25, Clothing – 96, Electronics – 147 = **268**;
- May 2023: Clothing – 115, Electronics – 79, Home – 23, Toys – 43 = **260**;
- June 2023: Books – 11, Clothing – 137, Electronics – 72, Home – 104, Toys – 28 = **352**

Now, I will show you for each product for each month on the next page:

Month	Product	Total Quantity
01.2023	Action Figure	11
01.2023	Biography	93
01.2023	Board Game	56
01.2023	Cookbook	71
01.2023	Headphones	3
01.2023	Laptop	67
01.2023	Novel	64
01.2023	Smartphone	8
02.2023	Action Figure	64
02.2023	Biography	166
02.2023	Chair	163
02.2023	Cookbook	78
02.2023	Headphones	61
02.2023	Laptop	54
02.2023	Smartphone	9
02.2023	T-Shirt	206
03.2023	Chair	13
03.2023	Novel	82
03.2023	T-Shirt	62
04.2023	Biography	6
04.2023	Cookbook	19
04.2023	Headphones	78
04.2023	Jeans	96
04.2023	Laptop	68
04.2023	Smartphone	1
05.2023	Jacket	2
05.2023	Jeans	46
05.2023	Lamp	23
05.2023	Puzzle	43
05.2023	Smartphone	79
05.2023	T-Shirt	67
06.2023	Board Game	28
06.2023	Jacket	82
06.2023	Jeans	15
06.2023	Lamp	76
06.2023	Novel	11
06.2023	Smartphone	72
06.2023	T-Shirt	40
06.2023	Table	28

January 2023: Action Figures (11), Biographies (93), Board Games (56), Cookbooks (71), Headphones (3), Laptops (67), Novels (64), Smartphones (8) = **373**;

February 2023: Action Figures (64), Biographies (166), Chairs (163), Cookbooks (78), Headphones (61), Laptops (54), Smartphones (9), T-Shirts (206) = **801**;

March 2023: Chairs (13), Novels (82), T-Shirts (62) = **157**;

April 2023: Biographies (6), Cookbooks (19), Headphones (78), Jeans (96), Laptops (68), Smartphones (1) = **268**;

May 2023: Jackets (2), Jeans (46), Lamps (23), Puzzles (43), Smartphones (79), T-Shirts (67) – **260**;

June 2023: Board games (28), Jackets (82), Jeans (15), Lamps (76), Novels (11), Smartphones (72), T-Shirts (40), Tables (28) = **352**

Okay. Now I am going to save all that data to another file.

This report was created by:



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Programs and platforms used:

Microsoft Word

Microsoft Excel

AI (for clarification of questions)